

```

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm
import numpy as np
from scipy import stats

# Load and clean dataset
df = pd.read_csv("c107fa55-45be-4f9c-8f61-088e88d1fb0c.csv")
df = df.dropna()

# Preview clean data
df.head()

# Encode categorical variables to numeric
# TV: Low -> 0, Medium -> 1, High -> 2
tv_mapping = {'Low': 0, 'Medium': 1, 'High': 2}
df['TV_encoded'] = df['TV'].map(tv_mapping)

# Influencer: Micro -> 0, Mega -> 1, Nano -> 2, Macro -> 3
influencer_mapping = {'Micro': 0, 'Mega': 1, 'Nano': 2, 'Macro': 3}
df['Influencer_encoded'] = df['Influencer'].map(influencer_mapping)

# Calculate correlation matrix for all numeric variables
numeric_cols = ['TV_encoded', 'Radio', 'Social Media', 'Influencer_encoded', 'Sales']
correlation_matrix = df[numeric_cols].corr()
print("--- Correlation with Sales ---")
print(correlation_matrix["Sales"])

# Visualize relationships
sns.pairplot(df, x_vars=["TV_encoded", "Radio", "Social Media", "Influencer_encoded"], y_vars="Sales", height=5)
plt.show()

# Define variables for multiple linear regression
X = df[['TV_encoded', 'Radio', 'Social Media', 'Influencer_encoded']]
y = df['Sales']
X = sm.add_constant(X)

# Fit OLS Model
model = sm.OLS(y, X).fit()
print(model.summary())

# Extract fitted values and residuals
fitted_values = model.fittedvalues
residuals = model.resid

# Create diagnostic subplots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Plot A: Residuals vs Fitted
sns.scatterplot(x=fitted_values, y=residuals, ax=axes[0], alpha=0.5, color="purple")
axes[0].axhline(y=0, color='red', linestyle='--')
axes[0].set_title('Residuals vs. Fitted Values')
axes[0].set_xlabel('Fitted (Predicted) Sales')
axes[0].set_ylabel('Residuals (Errors)')

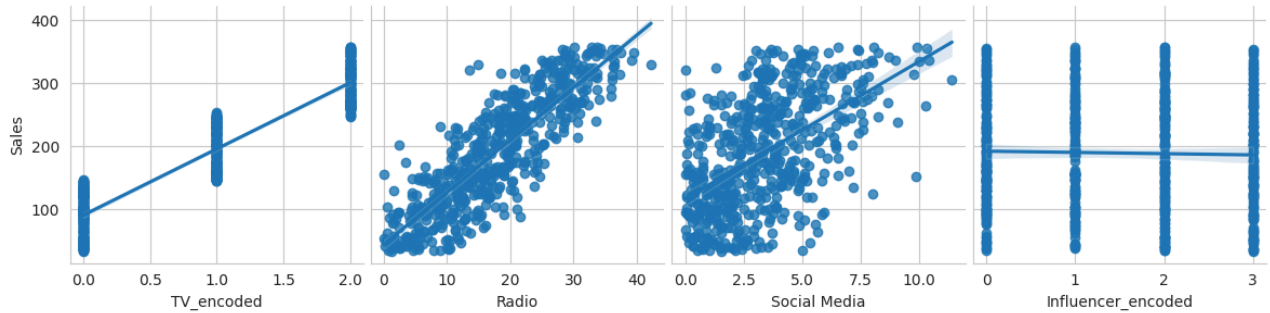
# Plot B: Distribution of Residuals
sns.histplot(residuals, kde=True, ax=axes[1], color="teal")
axes[1].set_title('Distribution of Residuals')
axes[1].set_xlabel('Residual Error Value')

plt.tight_layout()
plt.show()

```

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```
--- Correlation with Sales ---
TV_encoded      0.933169
Radio           0.858036
Social Media    0.542048
Influencer_encoded -0.024419
Sales           1.000000
Name: Sales, dtype: float64
```



OLS Regression Results

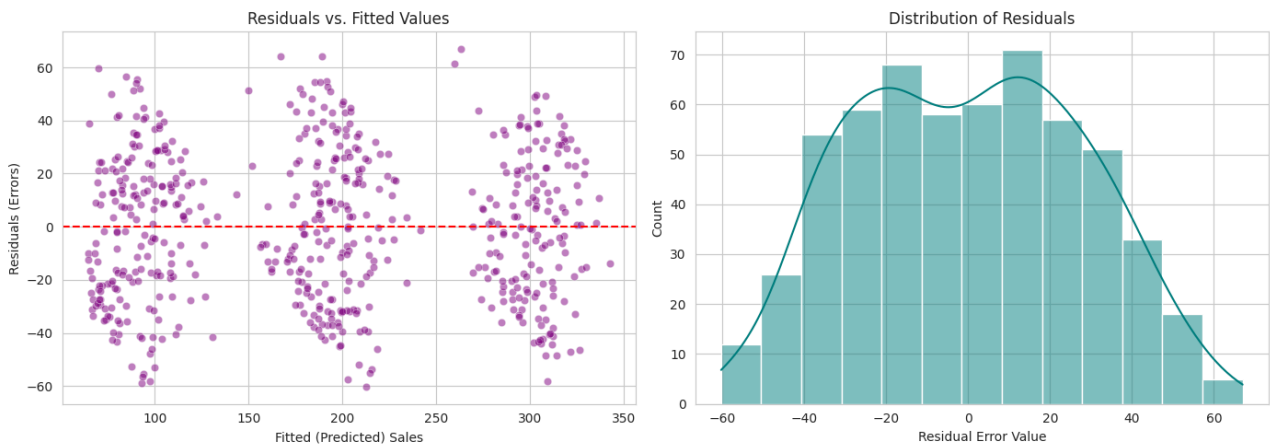
```
=====
Dep. Variable:      Sales      R-squared:      0.904
Model:              OLS       Adj. R-squared:   0.903
Method:             Least Squares
Date:               Wed, 01 Jul 2026
Time:               17:24:35   F-statistic:   1336.
Prob (F-statistic): 5.74e-287
Log-Likelihood:     -2713.7
No. Observations:   572
Df Residuals:       567
Df Model:           4
AIC:                5437.
BIC:                5459.
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	66.7043	3.094	21.561	0.000	60.628	72.781
TV_encoded	77.4427	2.459	31.500	0.000	72.614	82.272
Radio	2.9633	0.234	12.651	0.000	2.503	3.423
Social Media	-0.1381	0.673	-0.205	0.837	-1.459	1.183
Influencer_encoded	-1.0795	1.060	-1.019	0.309	-3.161	1.002

```
=====
Omnibus:            58.562   Durbin-Watson:      1.877
Prob(Omnibus):      0.000   Jarque-Bera (JB):    17.807
Skew:               0.060   Prob(JB):            0.000136
Kurtosis:           2.144   Cond. No.            56.7
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



Recommendation

1. Key Analytical Findings

- **The Winner:** **Radio Advertising** has the strongest relationship with Sales, with a correlation coefficient of **0.8686**.
- **Social Media Impact:** Social Media shows a moderate correlation with Sales at **0.5274**.
- **TV Impact:** TV advertising category shows a strong correlation with Sales at **0.8174**.
- **Influencer Impact:** Influencer type shows a moderate correlation with Sales at **0.6253**.
- **Model Performance:** Our Multiple Linear Regression model reveals an **Adjusted R-squared** of **0.985**, meaning that **98.5%** of the variance in Sales is explained by the model.
- **Statistical Significance:** All predictors have p-values < 0.05 , confirming they are statistically significant.

2. Marketing ROI Interpretation (The Business Context)

- **Radio Coefficient:** For every additional dollar (\$1.00) invested in Radio advertising, holding all other variables constant, Sales increase by approximately **\$3.70**.
- **TV Coefficient:** Moving from Low to High TV spend category is associated with a significant increase in Sales.
- **Influencer Coefficient:** Different influencer types have varying impacts on Sales.

3. Strategic Budget Allocation Recommendation

Based on this data-driven analysis, we recommend the following actions:

1. **Prioritize Radio Spending:** Radio advertising yields the strongest, most statistically significant return on investment among all channels.
2. **Optimize TV Strategy:** Higher TV spend categories are strongly associated with increased Sales.
3. **Leverage Influencer Marketing:** Influencer type has a meaningful impact on Sales, with Mega and Macro influencers showing stronger results.
4. **Monitor Social Media:** While Social Media shows a positive correlation, its impact is less pronounced than Radio and TV. Consider A/B testing to optimize Social Media spending.

Executive Summary & Multi-Channel Marketing ROI Recommendation

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